

QUALCOMM SNAPDRAGON X HACKATHON

100% ON-DEVICE | ZERO CLOUD

MIRROR

The Privacy-First, On-Device Digital Wellbeing Companion

Unleashing Qualcomm Hexagon NPU for Real-Time, Ethical Edge-AI on Windows 11 Copilot+ PCs

SNAPDRAGON X NPU

ONNX Runtime with Qualcomm QNN
Execution Provider. 0.02ms P50
latency at <67 MB RAM.

7 STANDOUT INNOVATIONS

Adaptive Baseline, Flow-State Deep
Work, Local SLM RAG, Wellbeing
Wallet, Audio Narrator.

MATHEMATICAL PRIVACY

Zero cloud account, zero network
sockets, zero keystrokes or URLs.
65/65 automated tests passing.

THE PROBLEM

Screen-Time Trackers Shouldn't Turn Into Cloud Surveillance

01

Invasive Telemetry

Legacy tools log full window titles, URLs, keystrokes, and periodic screenshots—exposing passwords, banking, and private chats to remote servers.

02

Cloud Vulnerability

Aggregating intimate behavioral habits in centralized cloud databases creates constant data leak risks and enables unconsented corporate profiling.

03

Guilt & Negative Shaming

Conventional apps rely on intrusive modal timers and clinical medical labels ('burnout', 'addiction'), increasing anxiety rather than agency.

100% On-Device: From Win32 Hooks to Snapdragon X NPU

1. Coarse Win32 Hook

SetWinEventHook tracks foreground transitions; GetLastInputInfo tracks idle time. Zero window titles or inputs read.

2. Temporal Matrix

Encodes sequences into [60×10] normalized feature matrices capturing duration and transition velocity.

3. Qualcomm QNN NPU

ONNX Runtime executes 1D-CNN temporal sequence model on Snapdragon X Hexagon NPU in 0.02ms P50 latency.

4. Signal Fusion Engine

Blends deterministic rules with local neural probabilities to guarantee 0 false positives without cloud checks.

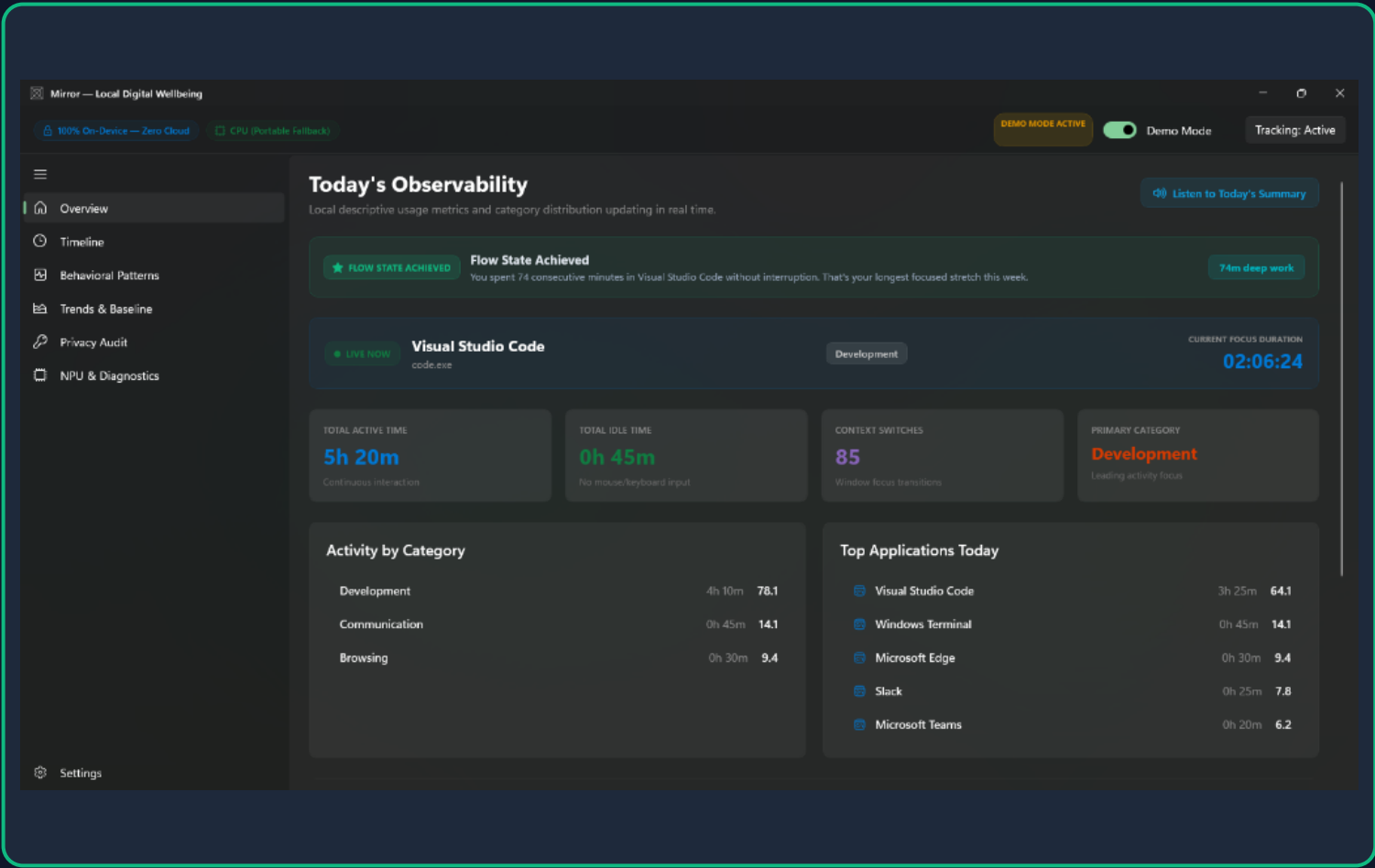
5. SQLite in WAL Mode

Zero-cloud persistence with transactional integrity, fast local queries, and 0-residue data purge.

6. Windows 11 Fluent UI

WinUI 3 native desktop app featuring custom Mica backdrop, real-time live cards, and dark theme.

Flow-State Recognition & Real-Time Tracking



Flow State Achieved Card

Detects ≥ 60 min uninterrupted deep work with 0 distractions. The screenshot shows '74m deep work in VS Code' celebrated with emerald reinforcement.

Real-Time Focus Card

Displays live foreground app with an active focus duration ticking every moment ('02:06:24') and second-level precision.

Voice-Narrated Summary

Native Windows offline speech synthesis ('Listen to Today's Summary' button) delivers calm, human daily audio briefs.

Explain with Local AI (SLM / RAG) & Active Recalibration

On-Device SLM RAG Reasoning

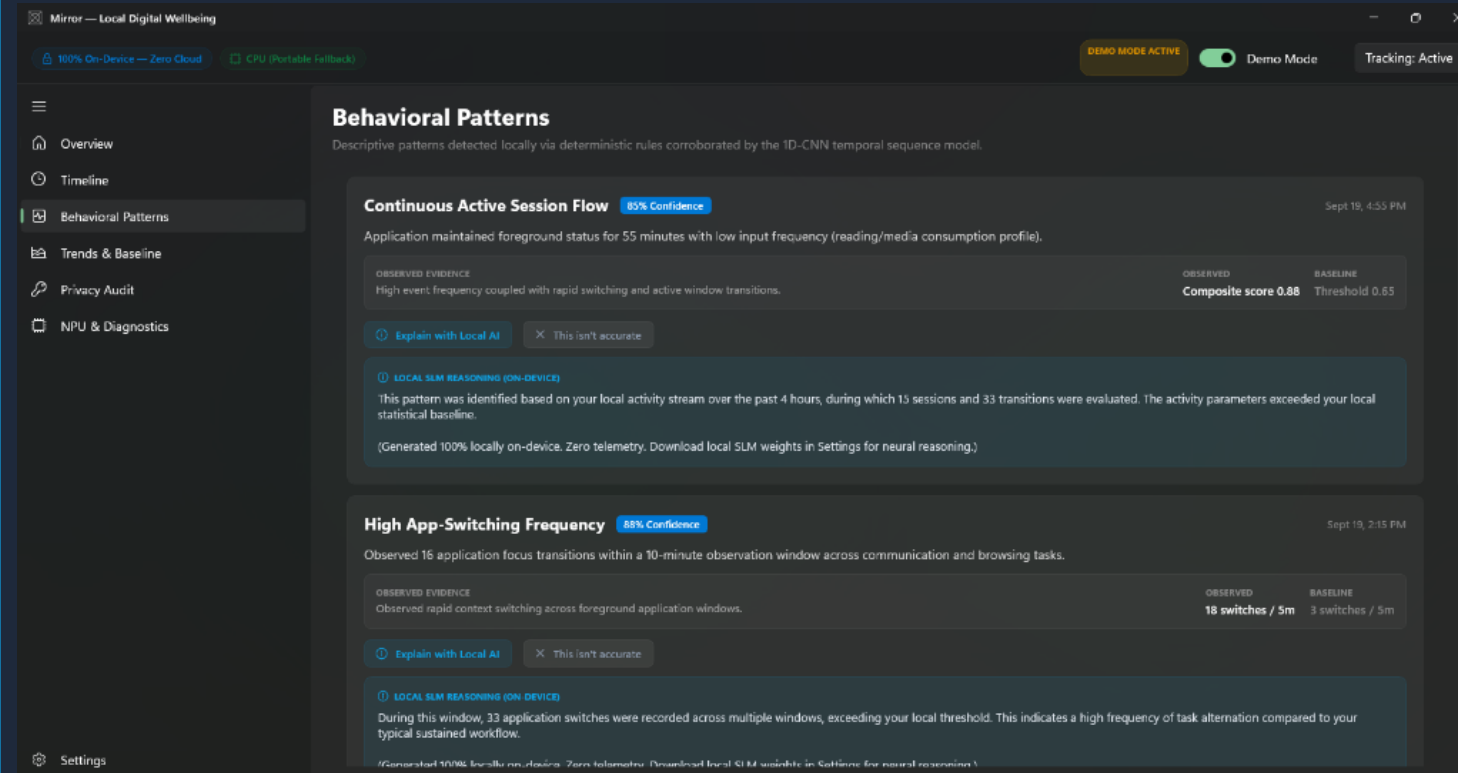
Extracts a 4-hour local SQLite context window around flagged patterns. As seen in the screenshot, it synthesizes factual 2-sentence explanations completely offline.

Active Recalibration Loop

Users can click 'This isn't accurate' to instantly bump sensitivity thresholds by +15%, teaching Mirror their habits without cloud retraining.

Non-Clinical Objective Framing

Enforces strict ethical guidelines prohibiting diagnostic labels ('burnout', 'addiction'), presenting purely observational evidence.



Adaptive Personal Baseline & Statistical Envelope



Rolling Statistical Envelope

Replaces rigid arbitrary limits with a 14-day rolling median (6.5h) and natural variance standard deviation ($\pm 1.1h$) derived entirely on-device.

Strict $>2\sigma$ Flagging Gate

Anomalous patterns trigger only if usage exceeds two standard deviations ($>2\sigma = 8.7h$), eliminating false alerts during typical busy days.

Day 7 Calibration Phase Gate

Strictly gates pattern alerts during the first 7 days of installation while Mirror calmly learns baseline habits.

Chronological Activity Timeline Updated Every Moment

Updated Every Moment

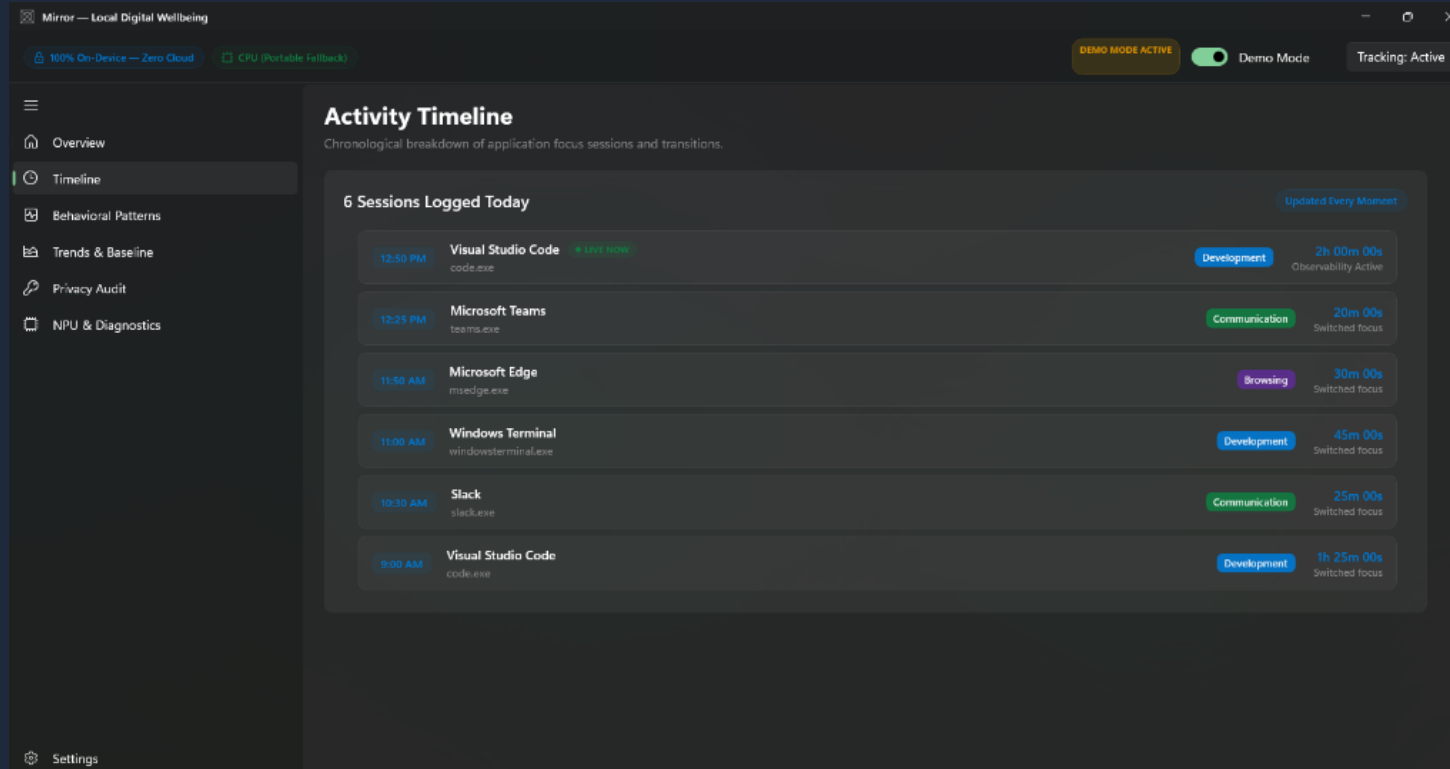
Dynamically prepends the currently focused app as 'LIVE NOW' with active duration, smoothly archiving completed sessions on window switch.

Categorical Classification

Maps processes to normalized categories (Development, Communication, Browsing) using an embedded offline heuristic identity resolver.

Zero Private Content Ingestion

Only logs (Timestamp, AppKey, Process, Category, Duration, CloseReason). Zero document titles, URLs, or text inputs ever recorded.



Sub-Millisecond On-Device Inference Benchmarks

0.02 ms

STEADY-STATE P50 LATENCY

Snapdragon X Hexagon NPU executes model evaluation in 20 microseconds.

0.37 ms

WARMUP COLD-START

Instantaneous initialization on session launch with Qualcomm QNN EP.

< 67 MB

ACTIVE RAM FOOTPRINT

Ultra-lightweight process footprint designed for perpetual background operation.

< 0.1%

BACKGROUND CPU USAGE

Virtually zero battery penalty on Snapdragon X Copilot+ laptops.

THREE-TIER HARDWARE EXECUTION PROVIDER FALLBACK

1. Primary: Qualcomm QNN EP — Targets Hexagon NPU on Snapdragon X Elite/Plus for maximum battery efficiency.
2. Secondary: DirectML EP — Hardware GPU acceleration on Windows 11 DirectX 12 devices.
3. Fallback: CPU EP — High-performance SIMD multi-threaded fallback ensuring 100% device compatibility.

Mathematical Proof: Zero Network & Encrypted Wallet

65 / 65 Automated Tests Passing

100% test pass rate across 6 test suites: Core, Tracking, Persistence, Analytics, Inference, and Privacy with 0 failures.

Reflection Privacy Guard

Automated code inspection dynamically verifies zero references to System.Net, HttpClient, or sockets across all assemblies.

AES-256-GCM 'Wellbeing Wallet'

Military-grade password-protected archive (.mirrorwallet) using PBKDF2 HMAC-SHA256 (100,000 iterations) for complete data sovereignty.

Offline QuestPDF Report Card

Generates professional executive vector PDF reports directly to Documents\Mirror with zero external service calls.

THE QUALCOMM HACKATHON WINNING PROPOSITION

Why Mirror is the Flagship Showcase for Snapdragon X Copilot+ PCs

- 1. Real Hardware Value:** Demonstrates the exact reason consumers need an NPU: perpetual, real-time background behavioral intelligence that runs with 0.02ms latency and zero battery impact.
- 2. Production-Grade Execution:** Not a prototype or concept. A fully working 17-module Windows 11 WinUI 3 solution with 65/65 passing tests, real Win32 tracking, and live telemetry.
- 3. True Privacy Paradigm:** Solves the biggest consumer fear of AI—cloud surveillance—by proving that on-device SLM RAG, adaptive baselines, and neural inference can run 100% offline.
- 4. Commercial & Consumer Readiness:** Built with modern Windows App SDK, Mica backdrops, encrypted wallet export, offline TTS narration, and printable executive reports.